

國立臺北大學 114 學年度第 2 學期期 末 考試試卷
National Taipei University Student's Answer Paper

系級/Department & Grade: 電機系2年級 科目/Course Title: 工程數學
(該科目所屬系級)/Course Given Department: _____
☐ 學士班 Bachelor Program ☐ 碩士班 Master Program ☐ 博士班 Ph.D. Program
學號/Student ID: _____ 姓名/Student Name: _____
任課教師/Instructor Name: 陳建宏 2026 年 Year 6 月 Month 9 日 Date

記 分 Score	教師簽章 Instructor Signature

1. $\vec{F} = xy\vec{a}_x - 2\vec{a}_y + 2\vec{a}_z$; $S: x^2 + y^2 = z^2, 0 \leq z \leq 1$, $\int_S \vec{F} \cdot d\vec{S} = ?$ (10%)

2. 座標轉換應用 (10%)

(a) $\iint_x \sin(x^2 + y^2) dx dy = ?$ $S: \{(x, y) | \pi \leq x^2 + y^2 \leq 9\pi\}$, (b) $\iiint_v \frac{1}{x^2 + y^2 + z^2} dx dy dz = ?$ $v: x^2 + y^2 + z^2 = 4$

3. $\vec{F} = -y\vec{a}_x + x\vec{a}_y - xyz\vec{a}_z$; $S: z = \sqrt{x^2 + y^2}, x^2 + y^2 \leq 9$, 利用 $\int_S (\nabla \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{l} = ?$ (10%)

4. $\oint_S x^2 dy dz - x^2 y dz dx + x^2 z dx dy = ?$, 封閉圓柱面外圍: $S: x^2 + y^2 = 9, 0 \leq z \leq 1$ (10%)

5. $\vec{F} = -\vec{a}_x + xyz\vec{a}_y - y^2\vec{a}_z$; $C: x = t, y = |t|, z = 1, t: -1 \rightarrow 1$, $\int_C \vec{F} \cdot d\vec{l} = ?$ (10%)

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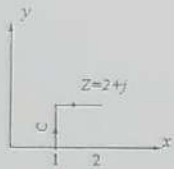
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6. Given Green 定理 $\leftarrow \frac{f(z)=u(x,y)+jv(x,y) \text{ 在封閉路徑 } C \text{ 的區域 } R \text{ 內可解析}}{\oint_C u(x,y)dx - v(x,y)dy = - \iint_R \left(\frac{\partial v(x,y)}{\partial x} + \frac{\partial u(x,y)}{\partial y} \right) dx dy} \rightarrow$ (10%)

$f(z) = e^z$, 由 Green 定理, 證明 $\oint_C f(z) dz = 0$, $C: |z|=1$

7. $\int_C z dz = ?$ (10%)



8. $f(z) = \frac{1}{z-1}$, $\oint_C f(z) dz = ?$, (a) $C_1: |z-2|=5$, (b) $C_2: |z-3|=1$; (c) $C_3 = C_4 + C_5$, C_4 (CCW): $|z-1|=1$, C_5 (CW): $|z-1|=2$ (10%)

9. $\oint_C \frac{3z}{(z+2)(z-1)^2} dz = ?$, (a) $C_1: |z+1|=0.5$, (b) C_2 : the boundary of the triangle with vertices 2, $2i$ and $-2i$. (10%)

10. $f(z) = \tan^{-1} z$ 的 Laurent 級數? (10%)